University of BRISTOL

COMS20018, Introduction to AI:

Philosophy of AI: Turing and Searle

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Agenda

We're going to look at two papers which are important in the history of the philosophy of AI:

1. *Computing Machinery and Intelligence* by Alan Turing, *Mind*, 59, 433-460, 1950.
2. *Minds, brains, and programs* by John Searle, *Behavioral and Brain Sciences* 3 (3): 417-457, 1980.

The link above to Searle's paper only takes you to a pre-print which does not contain critiques of Searle by other authors. To get the full paper (as a PDF), you need to go via the University library.

Asking "Can machines think?"

- Turing starts his paper as follows:

  > *I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think."*

This is a typical approach in philosophy: first consider what we even mean by the question we are investigating.

Not asking "Can machines think?"

- Turing quickly decides that the question "Can machines think?" is too ambiguous.
- He later says it is: "too meaningless to deserve discussion"
- The key innovation of his *Mind* paper is to **replace** the original question with another "which is closely related to it and is expressed in relatively unambiguous words."

The Imitation Game

- Turing imagines a game where a machine attempts to trick an interrogator into thinking it to be human.
- Turing calls it "The Imitation Game", later it became known as the "Turing Test".
- Turing's new question is: what will happen when a machine plays the imitation game?

What sort of machine?

- "we wish to exclude from the machines men born in the usual manner."
- "we only permit digital computers to take part in our game."
- Turing then goes on to demonstrate the universality of Turing machines (although he doesn't call them that!)

Turing's prediction on the Turing test

> *I believe that in about fifty years' time it will be possible, to programme computers, with a storage capacity of about 109, to make them play the imitation game so well that an average interrogator will not have more than 70 per cent chance of making the right identification after five minutes of questioning*

- OK, so had the 'Turing test' been passed in $\sim$ 2000?
- Has the 'Turing test' now been passed?

Turing's prediction on "educated opinion"

> *. . . I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.*

- Was he right?

Arguments against Turing machines thinking: Uncomputability

- Turing machines are limited: there are uncomputable problems, e.g. the Halting Problem
- But what about human limitations?

Arguments against Turing machines thinking: Discreteness

- Computers are discrete, whereas brains are continuous.
- A discrete machine can approximate continuous quantities to any desired degree of precision (consider floating point numbers)

Arguments against Turing machines thinking: Lovelace's objection

- Ada Lovelace: machines just do what we tell them to do.
- In the section on "Learning Machines", Turing says:

  Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain.

- Turing postulates a machine learning rules (symbolic learning) using a form of reinforcement learning.

Machine learning of natural language

It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. This process could follow the normal teaching of a child. Things would be pointed out and named, etc.

- How does this relate to how LLMs are trained?
- There's an argument here for *embodied AI*.

Searle's claim

1. "Intentionality in human beings (and animals) is a product of causal features of the brain."
2. "Instantiating a computer program is never by itself a sufficient condition of intentionality"

Also: "... *only* a machine could think, and only very special kinds of machines, namely brains and machines with internal causal powers equivalent to those of brains." and "no program by itself is sufficient for thinking".

The Chinese room

- Searle imagines himself locked in a room with 3 batches of Chinese writing.
- Searle is completely ignorant of the Chinese language (spoken or written) and cannot even distinguish Chinese characters from "meaningless squiggles".
- Unknown to Searle, the 3 batches are a "script", a "story" and some questions about the story.
- Searle is given instructions in English ("the program") that enable him to "correlate elements" of what's in the 3 batches of Chinese writing to generate answers (in Chinese) to the questions.

The Chinese room (continued)

Searle imagines that:

> *I get so good at following the instructions for manipulating the Chinese symbols and the programmers get so good at writing the programs that from the external point of view - that is, from the point of view of somebody outside the room in which I am locked - my answers to the questions are absolutely indistinguishable from those of native Chinese speakers. Nobody just looking at my answers can tell that I don't speak a word of Chinese*

and continues:

> *As far as the Chinese is concerned, I simply behave like a computer; I perform computational operations on formally specified elements. For the purposes of the Chinese, I am simply an instantiation of the computer program.*

Searle's main argument

. . . whatever purely formal principles you put into the computer, they will not be sufficient for understanding, since a human will be able to follow the formal principles without understanding anything.

Abelson's critique

the understanding is in the programmer's rule set, which the computer exercises

I would be willing to bet that by the Chinese symbol test, most of the people reading this don't really understand the transcendental number e, or economic inflation, or nuclear power plant safety, or how sailboats can sail upwind. (Be honest with yourself!)

At what point does a person graduate from "merely" manipulating rules to "really" understanding?

Block's critique

. . . his argument does not even have the right form, for in order to know whether we should reject the doctrine because of its alleged counterintuitive consequences, we must know what sort of evidence there is in favor of the doctrine. If the evidence for the doctrine is overwhelming, then incompatible intuitions should be ignored.

Does Searle really think that it is an initially congenial idea that a hunk of gray jelly is the seat of his intentionality? (Could one imagine a less likely candidate?) What makes gray jelly so intuitively satisfying to Searle is obviously his knowledge that brains are the seat of our intentionality. But here we see the difficulty in relying on considered intuitions, namely that they depend on our beliefs, and among the beliefs most likely to play a role in the case at hand are precisely our doctrines about whether the formal symbol-manipulation theory of thinking is true or false.

Dennett's critique

Dennett argues that the valid part of Searle's argument is not new:

> *"A computer whose only input and output was verbal would always be blind to the meaning of what was written" (Dennett 1969, p. 182). The idea has been around for a long time.*

Also:

> *Searle has confused different levels of explanation (and attribution). I understand English; my brain doesn't . . .*

If we suppose "the hero in the Chinese room" is more like a robot:

> *Are we to imagine that when he responds to the Chinese for "pass the salt, please" by getting his hand to grasp the salt and move it in a certain direction, he doesn't notice that this is what he is doing? In short, could anyone who became accomplished in this imagined exercise fail to become fluent in Chinese in the process?*